MAX PLANCK INSTITUTE
FOR DYNAMICS OF COMPLEX
TECHNICAL SYSTEMS
MAGDEBURG

DMP
DATA-DRIVEN MODELING OF
COMPLEX PHYSICAL SYSTEMS

Molecular Kinetics with Koopman Generators and Random Fourier Features

Feliks Nüske

July 15, 2025

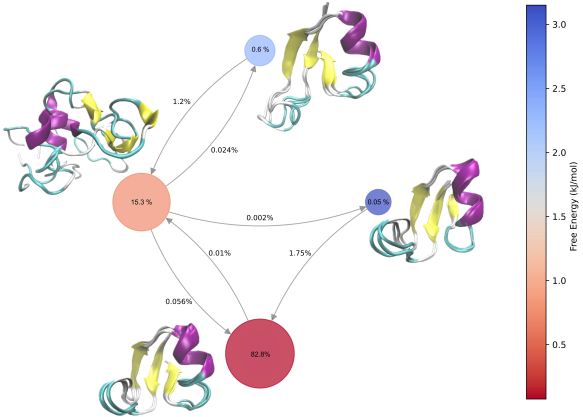
Outline

1. Metastability and Koopman Operators
2. Variational Approaches
3. Kernels and Random Features
4. Thermodynamic Interpolation
5. Kinetic Coarse Graining

joint work with: Cecilia Clementi, Weilong Chen, Boumediene Hamzi, Stefan Klus, Selma Moqvist, Vahid Nateghi, Frank Noé, Simon Olsson, Sebastian Peitz, Christof Schütte, Hao Wu

Metastability

- **Metastability** ⇔ systems with **long-lived states** under time evolution:



Operator Description

- **Koopman (adjoint) operator**

$$\mathcal{K}^t\phi(x) = \mathbb{E}\left[\phi(X_t) \,|\, X_0 = x\right].$$

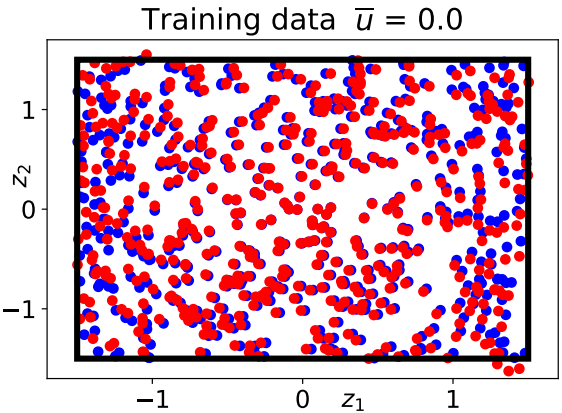
- Large t asymptotics:

$$\mathcal{K}^t\phi(x) \xrightarrow{t\to\infty} \mathbb{E}^\pi[\phi].$$

- Intermediate t:

$$\mathcal{K}^t\phi(x) = \mathbb{E}^\pi[\phi] + \sum_{i=1}^{r} e^{-\kappa_i t} \langle \phi, \psi_i \rangle_\pi \psi_i(x),$$

where $\kappa_i \ll 1$.



Koopman, P.N.A.S. (1931), **Mezic**, Nonlinear Dyn. (2005)

Spectral Analysis

- We need **dominant eigenpairs** of Koopman operator...

$$\mathcal{K}^t \psi_i = e^{-\kappa_i t} \psi_i, \qquad \kappa_i \ll 1.$$

Davies, J. London Math. Soc. (1982);
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Spectral Analysis

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- ... or lowest eigenpairs of the **generator**:

$$\mathcal{L} = \lim_{h \to 0} h^{-1} [\mathcal{K}^h - \mathrm{Id}], \qquad \mathcal{L}\psi_i = -\kappa_i \psi_i.$$

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- ... these encode long-lived states:

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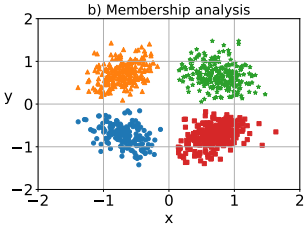

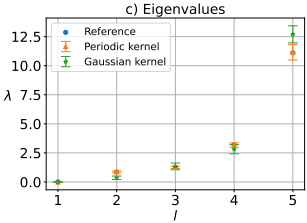

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Variational Approach (VAC)

True eigenfunctions maximize auto-correlation subject to orthogonality.

- Choose finitely many library functions:

$$\Psi(x) = \left[\psi_1(x) \cdots \psi_n(x)\right]^\top .$$

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FN, Keller, Pérez-Hernández, Mey, and Noé, J. Chem. Theory Comput. (2014);

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$$\mathbf{C}(t)\mathbf{v}_i = \hat{\lambda}_i(t)\mathbf{C}(0)\mathbf{v}_i, \qquad \mathbf{C}(t) = \mathbb{E}[\Psi(\mathbf{X}_s)\Psi(\mathbf{X}_{s+t})], \qquad \mathbf{C}(0) = \mathbb{E}[\Psi(\mathbf{X}_s)\Psi(\mathbf{X}_s)].$$

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- Variational Principle:

$$\sum_{i=0}^{r} \hat{\lambda}_i(t) \leq \sum_{i=0}^{r} \lambda_i(t) = \sum_{i=0}^{r} e^{-\kappa_i t}.$$

Noé and FN, SIAM MMS (2013);
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Generator Learning (gEDMD)

- Koopman generator is often known. For overdamped Langevin (Kolmogorov operator):

$$\mathcal{L}\psi = -\nabla V \cdot \nabla \psi + k_B T \Delta \psi.$$

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- Analogous **variational principle**:

$$\sum_{i=0}^{r} \hat{\kappa}_i \geq \sum_{i=0}^{r} \kappa_i.$$

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Generator Learning (gEDMD)

- ...only need basis functions and equilibrium samples:

$$\Psi(\mathbf{X}) = \begin{bmatrix} \psi_1(x_1) & \dots & \psi_1(x_m) \\ \vdots & \ddots & \vdots \\ \psi_n(x_1) & \dots & \psi_n(x_m) \end{bmatrix}, \qquad \mathcal{L}\Psi(\mathbf{X}) = \begin{bmatrix} \mathcal{L}\psi_1(x_1) & \dots & \mathcal{L}\psi_1(x_m) \\ \vdots & \ddots & \vdots \\ \mathcal{L}\psi_n(x_1) & \dots & \mathcal{L}\psi_n(x_m) \end{bmatrix},$$

- no trajectory integration.

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Choosing the Basis

- Many options: **TICA, MSM, VAMPnets, SRVs,...**

Schwantes and Pande, J. Chem Theory Comput. (2015), **Klus, Schuster, Muandet**, J. Nonlinear Science (2020), **FN, Klus, Hamzi**, Entropy (2020),

Choosing the Basis

- Many options: **TICA, MSM, VAMPnets, SRVs,...**
- Can also use a **kernel** to define the basis set:

$$\psi_i = k(x_i, \cdot).$$

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- Plug this into VAC equations:

$$\mathbf{K}_{\mathcal{L}\mathrm{X}}\mathbf{v}_i = \hat{\lambda}_i(t)\mathbf{K}_{\mathrm{XX}}\mathbf{v}_i, \qquad \mathbf{K}_{\mathrm{XX}}(r,s) = k(x_r, x_s), \qquad \mathbf{K}_{\mathcal{L}\mathrm{X}}(r,s) = \mathcal{L}k(x_r, x_s).$$

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- Matrices scale with data size. Hyper-parameter tuning can be expensive.
- Approximate using **random Fourier features**.

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Random Fourier Features

- For a **translation-invariant** kernels...

$$k(x, y) = \gamma(x - y).$$

Rahimi and Recht, Advances in Neural Information Processing Systems, 2007.

Random Fourier Features

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- ...write is as an inverse Fourier transform:

$$k(x, y) = \gamma(x - y) = (2\pi)^{-d/2} \int \hat{\gamma}(\omega) e^{i\omega^\top (x-y)} \, \mathrm{d}\omega.$$

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- If $k(x, x) = 1$, then $\mathrm{d}\rho = (2\pi)^{-d/2} \hat{\gamma}(\omega) \, \mathrm{d}\omega$ is a probability measure:

$$k(x, y) = \gamma(x - y) = \mathbb{E}^{\omega \sim \rho} \left[e^{i\omega^\top x} \left(e^{i\omega^\top y} \right)^* \right] \approx \frac{1}{p} \sum_{\ell=1}^{p} e^{i\omega_\ell{}^\top x} \left(e^{i\omega_\ell{}^\top y} \right)^*$$

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RFF and VAC

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Equivalent to VAC with random basis $\psi_l(x) = e^{i\omega_l^\top x}$.

FN and Klus, J. Chem. Phys., 2023

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Equivalent to VAC with random basis $\psi_l(x) = e^{i\omega_l{}^\top x}$.

- Prodecure:
- Choose a kernel k, determines ρ.
- Draw random frequencies $\omega_1, \ldots, \omega_p$ from ρ.
- Apply VAC with basis set

$$\psi_l(x) = \cos(\omega_l{}^\top x + b_l).$$

FN and Klus, J. Chem. Phys., 2023

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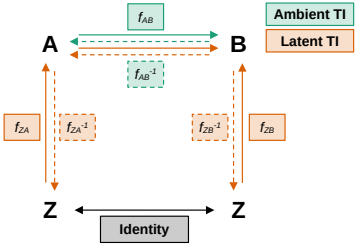
- Train on high-temperature data, extrapolate to target temperature.

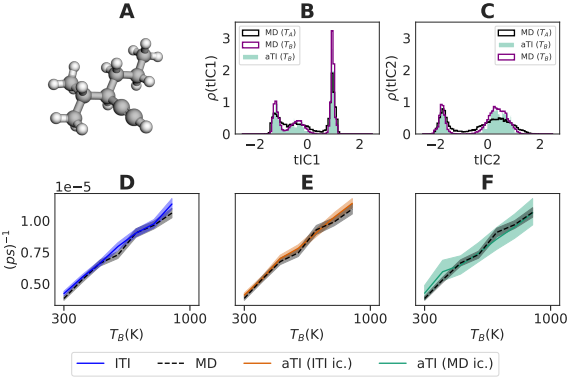


Dibak, Klein, Krämer, Noé, Physical Review Research 2022,

Moqvist, Chen, Schreiner, FN and Olsson, J. Chem. Theory Comput., 2025

Thermodynamic Interpolation

- Use **temperature-dependent** normalizing flow with stochastic interpolant.
- Train on high-temperature data, extrapolate to target temperature.
- Learn gEDMD model with random Fourier features on extrapolated data.
- Extract **temperature dependent transition rates / timescales**.



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Coarse Grained Generator

- **Coarse-graining map** $\xi : \mathbb{R}^d \mapsto \mathbb{R}^n$, $n \leq d$.
- Apply Zwanzig projector $\mathcal{L}^\xi = \mathcal{P}\mathcal{L}\mathcal{P}$; generates an **effective SDE**.

Legoll and Lelièvre, Nonlinearity (2010),
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- Fundamental identity relating projected and full generator:

$$\left\langle \psi_i \circ \xi,\, \mathcal{L}(\psi_j \circ \xi)\, e^{-\beta V} \right\rangle = \left\langle \psi_i,\, \mathcal{L}^\xi \psi_j\, e^{-\beta F} \right\rangle .$$

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- gEDMD with **basis function on CG space** approximates both $\mathcal{L}$ and $\mathcal{L}^\xi$.
- ... can do this using only full-state data.

Legoll and Lelièvre, Nonlinearity (2010),
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Learning CG Dynamics

- CG dynamics require **effective potential** F and **effective diffusion** a^{ξ}:

Nateghi and FN, J. Chem. Theory Comput. (2025, in press), *arxiv 2409.16396*

Learning CG Dynamics

- CG dynamics require **effective potential** F and **effective diffusion** a^ξ:
- Learning problems for F and a^ξ can be formulated: **force matching** and **diffusion learning**.

$$a^\xi = \arg\min_{a^\theta} \frac{1}{m} \sum_{i=1}^{m} \left\| a^\theta(\xi(x_i)) - a^\xi_{\mathrm{loc}}(x_i) \right\|_F^2 ,$$

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- **gEDMD model** for CG dynamics can be directly computed without trajectory integration:

$$\left\langle \psi_r, \mathcal{L}^\theta \psi_s \, e^{-\beta F} \right\rangle_\nu = -\frac{1}{2} \int \nabla \psi_r(z) a^\theta(z) \nabla \psi_s(z)^\top \, e^{-\beta F(z)} \, \mathrm{d}z.$$

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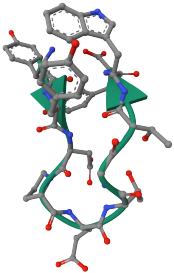
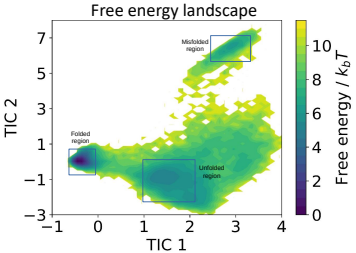
Learning CG Dynamics

- Coarse-graining of Chignolin mini-protein.
- Project into 2d TICA space; minimal effect of Markovian approx.





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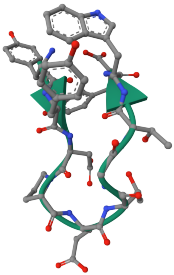

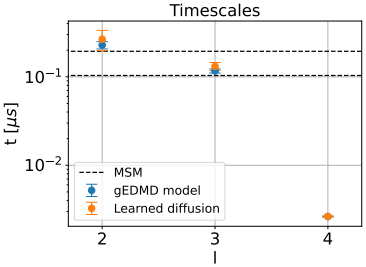

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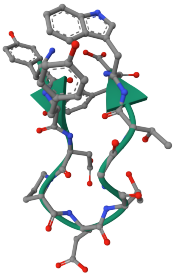
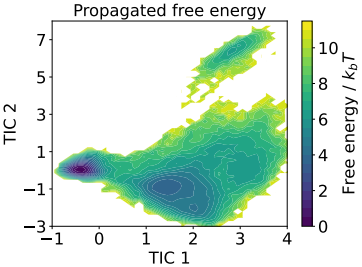
- Leads to 2d-surrogate model that re-covers slow processes of full protein simulation.
- Run simulations in TICA space; recover free energy.





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Thank you!

References:

- **Noé and FN**, A variational approach to modeling slow processes in stochastic dynamical systems, SIAM MMS, 2013.
- **Klus, FN, Peitz, Niemann, Clementi, and Schütte**, Data-driven approximation of the Koopman generator: Model reduction, system identification, and control, Physica D, 2020
- **FN and Klus**, Efficient Approximation of Molecular Kinetics using Random Fourier Features, Journal of Chemical Physics 159, 074105, 2023
- **Moqvist, Chen, Schreiner, FN, and Olsson**, Thermodynamic Interpolation: A generative approach to molecular thermodynamics and kinetics, Journal of Chemical Theory and Computation, 21(5):2535–2545, 2025
- **Nateghi and FN**, Kinetically Consistent Coarse Graining using Kernel-based Extended Dynamic Mode Decomposition, J. Chemical Theory Computation (2025, in press), arxiv 2409.16396.